

ENV 4120

Air Pollution and Hazardous Waste Control

Today's Topics

- Air pollution sources, transport and effects
- Expectation
 - Familiar with source types of air pollution
 - Know the six CAPs and their major sources

Sources of air pollution

Emission sources

- Primary vs secondary
 - Directly emitted vs formed in the air
- Natural vs man-made (anthropogenic)
- Stationary vs mobile
 - Stationary stacks, moving cars etc...
- And also by shapes
 - Point, nonpoint, area, volume, line etc....

Natural Sources

Volcano



USGS image by C. Newhall - September, 1984 (www.geology.com)

Sea



Credit: Photo by Michele Hogan
<http://www.sciencedaily.com/releases/2007/09/070919175542.htm>

Forest



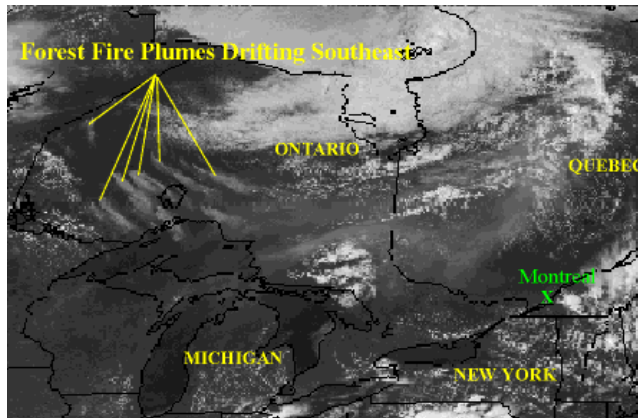
www.ivm.vu.nl/en/projects/Departments/economics/Reduced_Emission_from_Deforestation_and_Degradation_in_the_Southern_Cardamom_Ecosystem_Cambodia/index.asp

Dust storm



NOAA Photo Library, Historic NWS collection

Forest fire



Satellite image for June 14, 1996, at 1545 EST
(www.radar.mcgill.ca/forest_fire.html)

Hot springs



www.phase2.net/claygraphics/Images/bruce%20images/Tahcharian%20Hot%20Springs.jpg

Anthropogenic Sources

Power plants



(David Klobucar/Associated Press)
<http://www.cbc.ca/news/canadavotes/story/2008/09/19/f-carbontaxprimer.html>

Automobile



www.hybridmile.com/news/obama-supports-californi/

Industry



pollutionarticles.blogspot.com/2009/02/carbon-emissions-causing-huge-ocean.html

Lawn care



www.re-nest.com/uimages/re-nest/mowerwoody052509.jpg

Farms



www.extension.iastate.edu/Pages/communications/EPC/Su02/emissions.html

Land fill



Landfill <http://www.jber.af.mil/shared/media/photodb/photos/2012/08/120821-F-BP133-050.jpg>

CAPs and HAPs

CAPs & HAPs

- 6 Criteria Air Pollutants (CAPs):
 - O₃, CO, NO₂, SO₂, Pb, PM (PM_{2.5} & PM₁₀)
 - Concentration level set in NAAQS (in CAA)
 - Ubiquitous, most harmful to health and environment
- 189 Hazardous Air Pollutants (HAPs)
 - A.K.A. toxic air pollutants or air toxics
 - Known/suspected to cause cancer, or other serious health or environmental effects
 - Goal: reduce emissions (NESHAPS)
 - Primary, man-made emissions dominates

Particulate Matter (PM)

- A.K.A. aerosol, can be a solid/liquid/mixed
- PM_{10} & $PM_{2.5}$
 - PMs with aerodynamic diameter < 10 or $2.5 \mu m$
 - $1 \mu m = 1 \times 10^{-6} m$
 - Can lodge in airway once inhaled.
 - $PM_{2.5}$ penetrate deep into lungs (alveoli region)
- Many emission sources
 - Fire, dust, agriculture, fuel combustion, cars etc...
 - Both primary/secondary, natural/man-made sources
 - Secondary organic aerosol (SOA) is a major research topic

Nitrogen Dioxide (NO₂)

- Usually NO and NO₂ are measured together
 - NO eventually oxidized to NO₂
 - NO + NO₂ = NO_x (different from NO_y and NO_z!)
 - NO_z = a combination of several reactive nitrogen species
 - NO_y = NO_x + NO_z
- Mostly man-made, primary emissions (fuel combustion at high temperatures)
 - Mobile source, power plants dominates
- Some natural sources
 - Lightning generate NO_x
 - Some plants also emit NO_x

Ground-level ozone (O_3)

- One major component of photochemical smog
- Ground-level O_3 BAD; stratospheric O_3 GOOD.
 - Health effect when exposed near ground
 - Stratospheric O_3 layer absorbs harmful UV
- Secondary pollutants, negligible direct emissions
 - Primarily from NO_x and VOC, mixing with sunlight
 - VOC: Volatile Organic Compounds
 - From evaporation of organic solvents, incomplete combustion etc...
 - Trees emits VOC (biogenic): mostly isoprene
 - Globally, trees emits more VOC than man-made sources
 - Ronald Reagan (1981): Trees cause more pollution than automobiles do

Sulfur Dioxide (SO₂)

- From combustion of sulfur-containing fuel
- Primary, anthropogenic sources dominate
 - Coal fire power plants is the biggest source
 - Also industry, mobile, fire (wild + prescribed)
 - Some natural source: sulfur springs
- Major contributor to acid rain, especially northeastern US

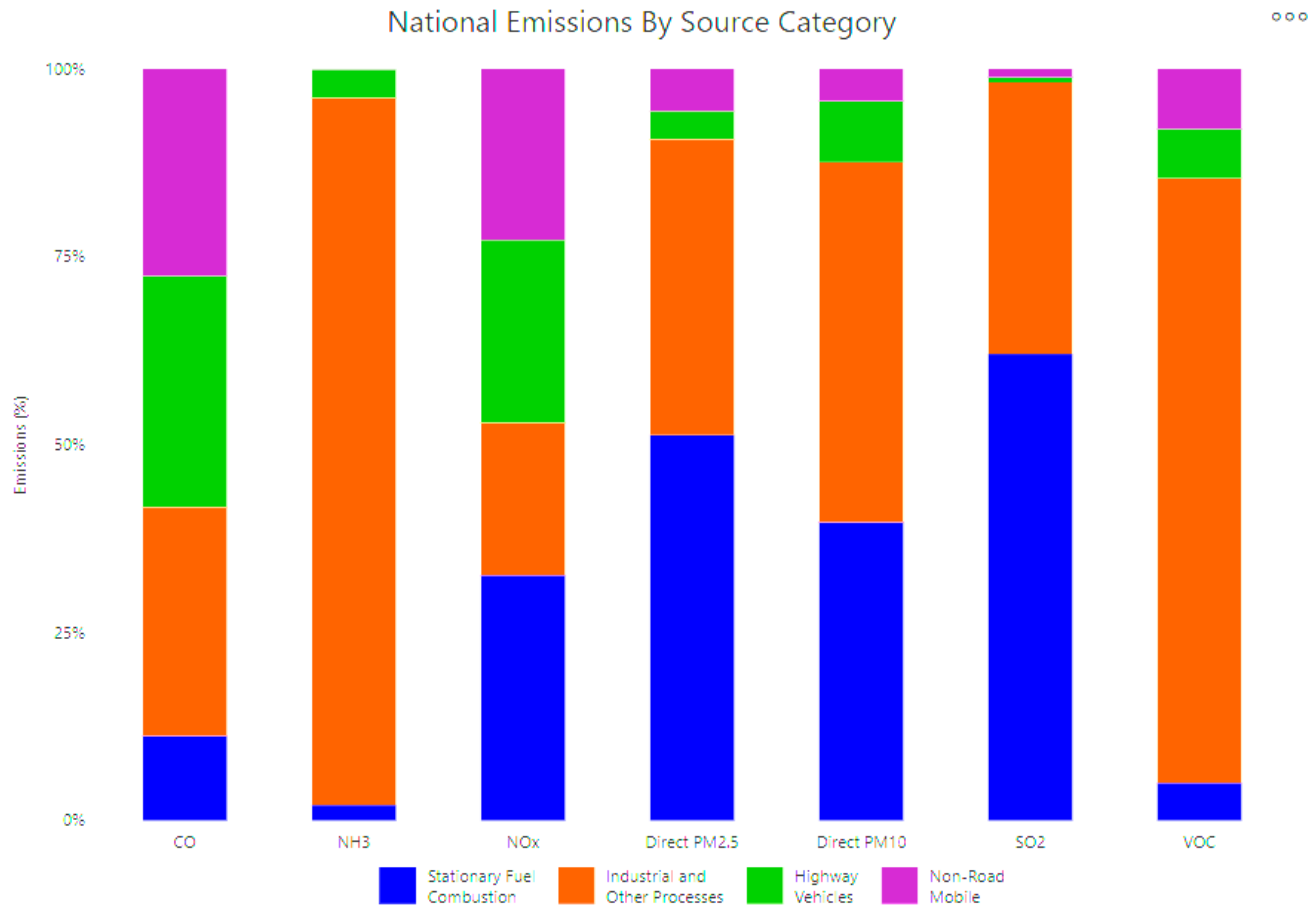
Carbon Monoxide (CO)

- Colorless, odorless, and tasteless gas
- Mostly primary, from incomplete fuel combustion
 - Mobile source, fire are the biggest ones
- Some plants emits CO (biogenic)
- CO binds with hemoglobin, reduce oxygen carrying capacity of blood

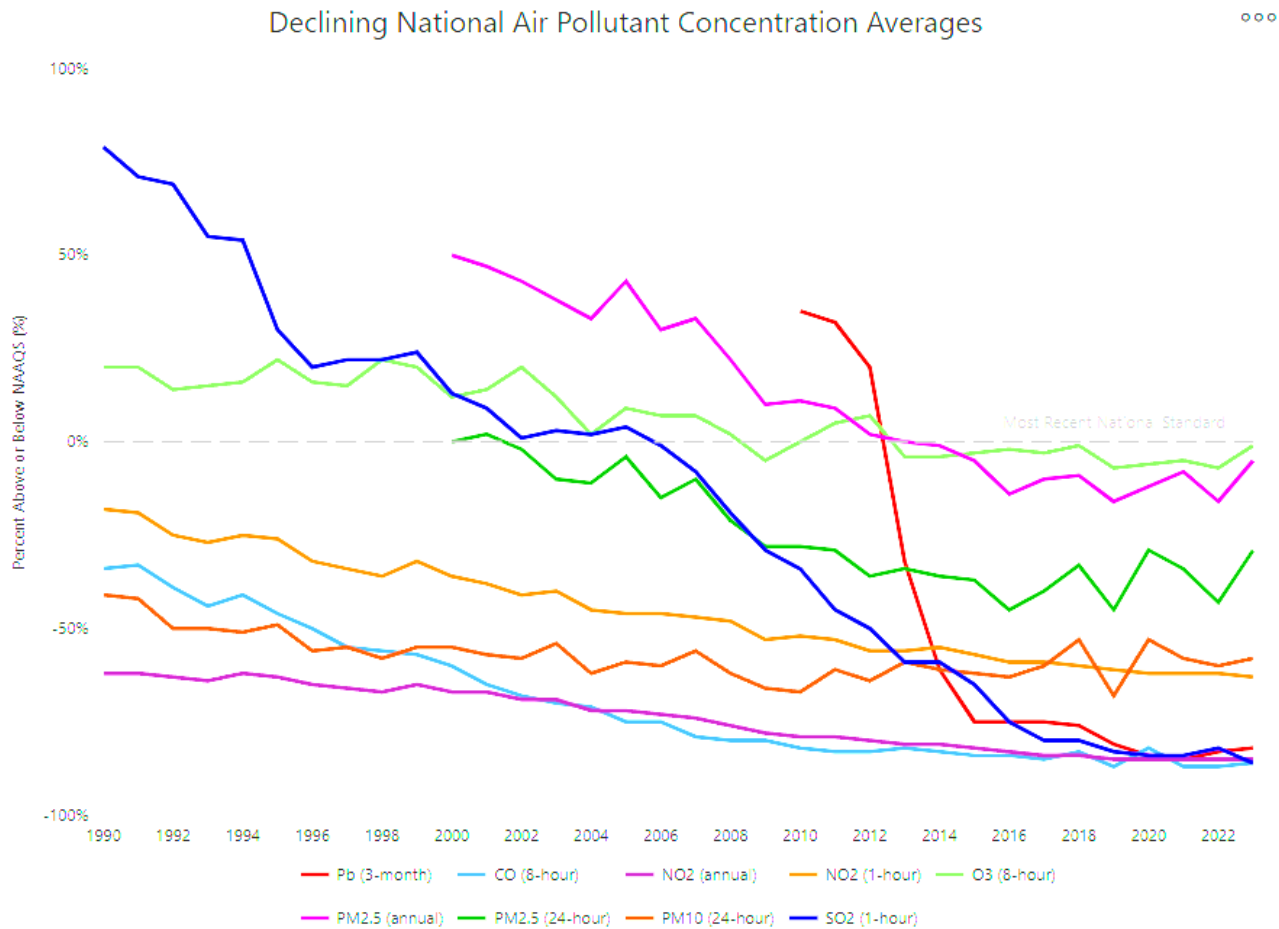
Lead (Pb)

- Particulate lead
- Combustion of fuels with lead additives
- Leaded gas phased out → lead in ambient air decreased by 99%
 - Leaded gas invented by Thomas Midgley Jr. who also brought us Freon (CFC)
- Lead particularly toxic to children
- Houses built before 1978 has lead paint

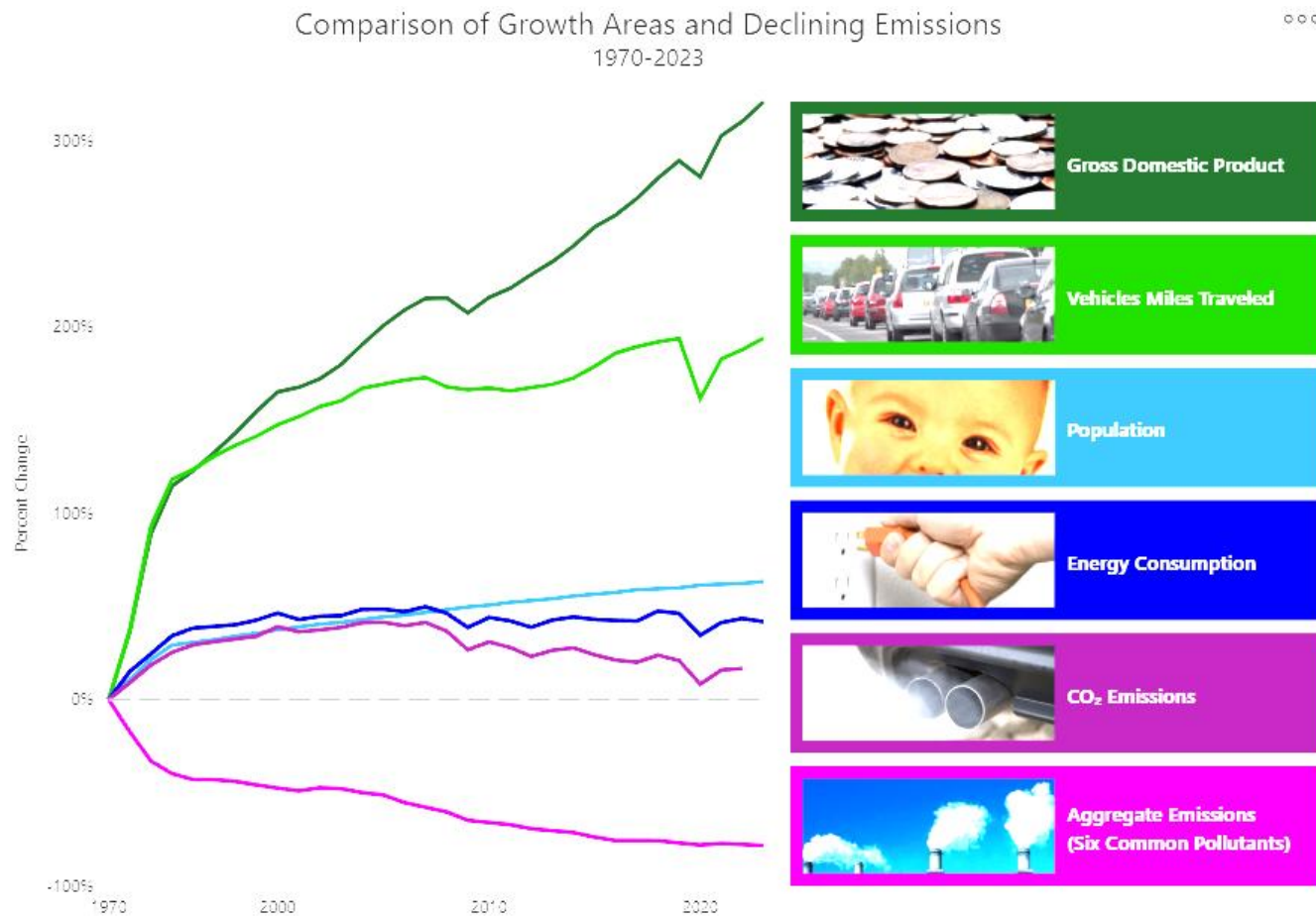
Emission sources



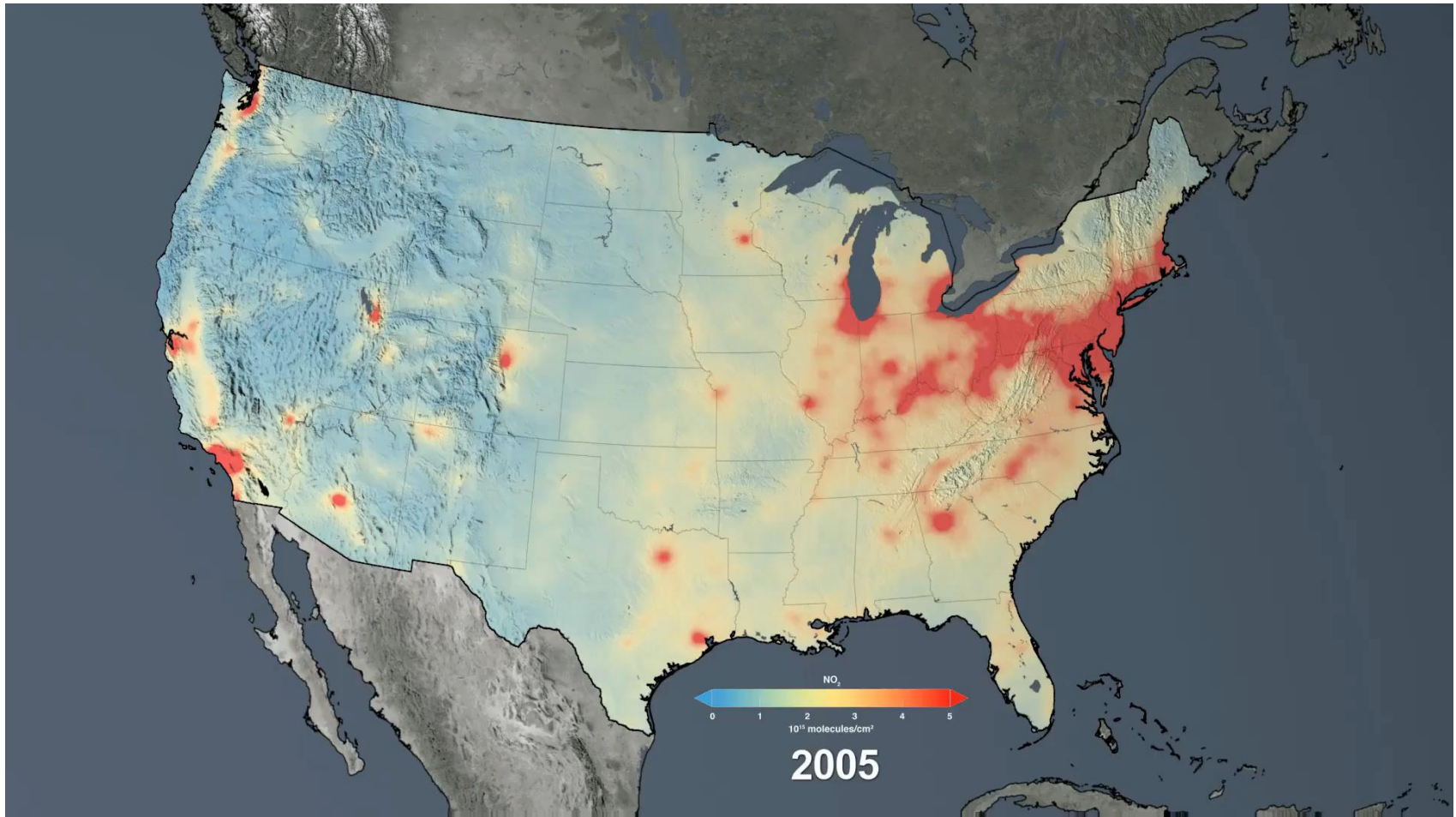
Concentration trend in the US



Emission trend in the US



NASA satellite NO₂ concentration



More information on trend of CAPs see:
<https://gispub.epa.gov/air/trendsreport/2024>
Also see here: <https://airquality.gsfc.nasa.gov/>

NASA satellite SO₂ concentration

